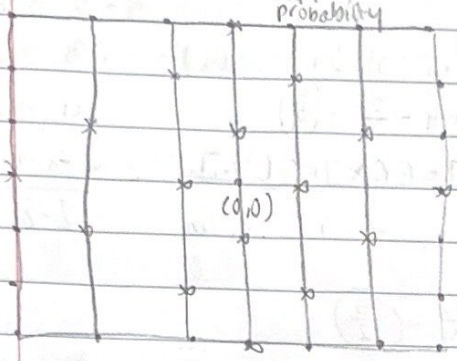
✓ ✓ ✓ ✓ ✓

X 2 3 4 5 6

HW 7

Due 11/22

1) 1. possible combinations



$\frac{9.4+3.2+4.1}{64}$

$3,0 = \frac{1}{64}$

$2,1 = \frac{3}{64}$

$1,0 = \frac{9}{64}$

R RL $\frac{31}{2} = 3$

R VD 31 = 6

count prob of ending at each spot, some are 0

$f(\pm 3,0) = f(0,\pm 3) = \frac{1}{64}$

$f(\pm 1,\pm 2) = f(\pm 2,\pm 1) = \frac{3}{64}$

$f(\pm 1,0) = f(0,\pm 1) = \frac{9}{64}$

$f(0,0) = f(\pm 1,\pm 1) = f(\pm 1,\pm 3) =$

$f(\pm 3,\pm 1) = f(\pm 2,0) = f(0,\pm 2)$

$= f(\pm 2,\pm 2) = f(\pm 2,\pm 3) = f(\pm 3,\pm 2) =$

$f(\pm 3,\pm 3) = 0$

$$f(S,T) = \begin{cases} \frac{1}{64}, & (S,T) = (\pm 3,0), (0,\pm 3) \\ \frac{3}{64}, & (S,T) = (\pm 1,\pm 2), (\pm 2,\pm 1) \\ \frac{9}{64}, & (S,T) = (\pm 1,0), (0,\pm 1) \\ 0, & (S,T) = (0,0), (\pm 1,\pm 1), (\pm 1,\pm 3), (\pm 3,\pm 1), (\pm 2,0), (0,\pm 2), \\ & (\pm 2,\pm 2), (\pm 2,\pm 3), (\pm 3,\pm 2), (\pm 3,\pm 3) \end{cases}$$

2. X now 0-6 Y now 0-6

$S=-3$ $P_X(0) = 0+0+0+\frac{1}{64}+0+0+0 = \frac{1}{64}$

$S=-2$ $P_X(1) = 0+0+\frac{3}{64}+0+\frac{3}{64}+0+0 = \frac{6}{64} = \frac{3}{32}$

$S=-1$ $P_X(2) = 0+\frac{3}{64}+0+\frac{9}{64}+0+\frac{3}{64}+0 = \frac{15}{64}$

$S=0$ $P_X(3) = \frac{1}{64}+0+\frac{9}{64}+0+\frac{9}{64}+0+\frac{1}{64} = \frac{20}{64} = \frac{10}{32}$

$S=1$ $P_X(4) = 0+\frac{3}{64}+0+\frac{9}{64}+0+\frac{3}{64}+0 = \frac{15}{64}$

$S=2$ $P_X(5) = 0+0+\frac{3}{64}+0+\frac{3}{64}+0+0 = \frac{6}{64} = \frac{3}{32}$

$S=3$ $P_X(6) = 0+0+0+\frac{1}{64}+0+0+0 = \frac{1}{64}$

$T=-3$ $P_Y(0) = 0+0+0+\frac{1}{64}+0+0+0 = \frac{1}{64}$

$T=-2$ $P_Y(1) = 0+0+\frac{3}{64}+0+\frac{3}{64}+0+0 = \frac{6}{64} = \frac{3}{32}$

$T=-1$ $P_Y(2) = 0+\frac{3}{64}+0+\frac{9}{64}+0+\frac{3}{64}+0 = \frac{15}{64}$

$T=0$ $P_Y(3) = \frac{1}{64}+0+\frac{9}{64}+0+\frac{9}{64}+0+\frac{1}{64} = \frac{20}{64} = \frac{10}{32}$

$T=1$ $P_Y(4) = 0+\frac{3}{64}+0+\frac{9}{64}+0+\frac{3}{64}+0 = \frac{15}{64}$

$T=2$ $P_Y(5) = 0+0+\frac{3}{64}+0+\frac{3}{64}+0+0 = \frac{6}{64} = \frac{3}{32}$

$T=3$ $P_Y(6) = 0+0+0+\frac{1}{64}+0+0+0 = \frac{1}{64}$

2) 1. x = # good items — $\frac{6}{10}$ — 15 items total

y = # "second" items — $\frac{3}{10}$ — $\left(\frac{6}{10}\right)^x \left(\frac{3}{10}\right)^y \left(\frac{1}{10}\right)^z$

z = $15-x-y$ = # defective items — $\frac{1}{10}$ — ways to pick: $\frac{15!}{x!\,y!\,15-x-y!}$

$$f(x,y) = \frac{15!}{x!\,y!\,15-x-y!} \cdot \left(\frac{6}{10}\right)^x \cdot \left(\frac{3}{10}\right)^y \cdot \left(\frac{1}{10}\right)^{15-x-y}$$

2. $P(X=10, Y=4) = \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11}{10!\,4!\,1!} \cdot \left(\frac{6}{10}\right)^{10} \cdot \left(\frac{3}{10}\right)^4 \cdot \left(\frac{1}{10}\right)^1 = \frac{15015 \cdot 6^{10} \cdot 3^4}{10^{15}} = 0.0735$

3. $f_x(x) = \sum_{y=0}^{15-x} \frac{15!}{x!\,y!\,(15-x-y)!} (0.6)^x (0.3)^y (0.1)^{15-x-y}$

$= (0.6)^x \frac{15!}{x!} \sum_{y=0}^{15-x} \frac{1}{y!\,(15-x-y)!} (0.3)^y (0.1)^{15-x-y}$

$15-x!$ $\binom{15-x}{y}$

cancels $15-x!\;y!\,15-x-y!$

$\binom{15}{x}\left(\frac{6}{10}\right)^x \left(\frac{3}{10}+\frac{1}{10}\right)^{15-x}$

3. $f_x(x) = \binom{15}{x}\left(\frac{6}{10}\right)^x \left(\frac{4}{10}\right)^{15-x}$

$\binom{15}{x}$: ways to order items; $\left(\frac{6}{10}\right)^x$: prob of picking x good items; $\left(\frac{4}{10}\right)^{15-x}$: prob of picking $y+z$ (or $15-x$) "second" or defective items

4. $P(X \le 11) = 1 - (P(X=12) + P(X=13) + P(X=14) + P(X=15)) =$

$\binom{15}{12}\left(\frac{6}{10}\right)^{12}\left(\frac{4}{10}\right)^3 + \binom{15}{13}\left(\frac{6}{10}\right)^{13}\left(\frac{4}{10}\right)^2 + \binom{15}{14}\left(\frac{6}{10}\right)^{14}\left(\frac{4}{10}\right)^1 + \binom{15}{15}\left(\frac{6}{10}\right)^{15}\left(\frac{4}{10}\right)^0$

$0.06339 + 0.02194 + 0.0047 + 0.00047$

$1 - 0.0905 = 0.9095$

3) $\mu_X = \sum_{x=\{1,2,3,4\}} x \cdot (P(X=x)) = 1\cdot\frac{1}{4} + 2\cdot\frac{1}{4} + 3\cdot\frac{1}{4} + 4\cdot\frac{1}{4} = \frac{10}{4} = \boxed{\frac{5}{2}}$

$\sigma_X^2 = E[X^2] - (E[X])^2 = 1^2\cdot\frac{1}{4} + 2^2\cdot\frac{1}{4} + 3^2\cdot\frac{1}{4} + 4^2\cdot\frac{1}{4} - (\frac{5}{2})^2 =$

$\frac{1}{4} + 1 + \frac{9}{4} + 4 - \frac{25}{4} = \boxed{\frac{5}{4}}$

Z = outcome of die 2 $\quad E[Y] = E[X] + E[Z]$

so, $\mu_Y = \frac{5}{2} + \frac{5}{2} = \boxed{5}$

$\sigma_Y^2 = \sigma_X^2 + \sigma_Z^2 = \frac{5}{4} + \frac{5}{4} = \boxed{\frac{5}{2}}$

$x = 1,2,3,4$

$y = 2,3,4,5$

$3,4,5,6$

$4,5,6,7$

$5,6,7,8$

2-8

$COV(X,Y) = E[XY] - \mu_X\mu_Y$ $\quad \frac{15}{2}$ from earlier ↓

$E[XY] = E[X(X+Z)] = E[X^2 + XZ] = E[X^2] + E[XZ] =$

$\frac{15}{2} + (E[X]\cdot E[Z]) = \frac{15}{2} + \frac{5}{2}\cdot\frac{5}{2} = \frac{15}{2} + \frac{25}{4} = \frac{55}{4}$

$COV(X,Y) = \frac{55}{4} - (\frac{5}{2})(5) = \frac{110}{8} - \frac{100}{8} = \frac{10}{8} = \boxed{\frac{5}{4}}$

$\rho = \frac{\sigma_{XY}}{\sigma_X\sigma_Y} = \frac{\frac{5}{4}}{\sqrt{\frac{5}{4}}\sqrt{\frac{5}{2}}} = \frac{\frac{5}{4}}{\frac{5}{\sqrt{8}}} = \frac{5}{4}\cdot\frac{\sqrt{8}}{5} = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2} \approx \boxed{0.71}$

4) 1. $P(0,0) = \frac{1}{4} \quad f_X(0) = \frac{1}{4} \quad f_Y(0) = \frac{1}{2}$

$\frac{1}{4} \neq \frac{1}{4}\cdot\frac{1}{2} = \frac{1}{8}$ since $P(0,0) \neq f_X(0)\cdot f_Y(0)$, X and Y are not independent.

2. $COV(X,Y) = E[XY] - \mu_X\mu_Y$

$\mu_X = 0\cdot\frac{1}{4} + 1\cdot\frac{1}{2} + 2\cdot\frac{1}{4} = 0 + \frac{1}{2} + \frac{1}{2} = \boxed{1}$ $\quad \mu_Y = 0\cdot\frac{1}{2} + 1\cdot\frac{1}{4} + -1\cdot\frac{1}{4} = \boxed{0}$

$E[XY] = 0\cdot0\cdot\frac{1}{4} + 1\cdot1\cdot\frac{1}{4} + 1\cdot-1\cdot\frac{1}{4} + 2\cdot0\cdot\frac{1}{4} = 0 + \frac{1}{4} - \frac{1}{4} + 0 = \boxed{0}$

$COV(X,Y) = 0 - (1\cdot0) = \boxed{0}$

$\rho = \frac{COV(X,Y)}{\sigma_X\sigma_Y} = \boxed{0}$ (from above)

HW 7 (cont.)

5) 1. $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{\frac{x+y}{32}}{\frac{3+2y}{32}} = \boxed{\frac{x+y}{3+2y}}$

$f_Y(y) = \sum_{x=1}^{2} \frac{x+y}{32} = \frac{1+y+2+y}{32} = \frac{3+2y}{32}$

$f_Y(1) = 5/32$ $f_Y(3) = 9/32$

$f_Y(2) = 7/32$ $f_Y(4) = 11/32$

2. $f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \boxed{\frac{x+y}{4x+10}}$

$f_X(x) = \sum_{y=1}^{4} \frac{x+y}{32} = \frac{x+1+x+2+x+3+x+4}{32} = \frac{4x+10}{32}$

3. $P(1 \le Y \le 3 | X=1) = f_{Y|X}(1|1) + f_{Y|X}(2|1) + f_{Y|X}(3|1)$

$= \frac{1}{7} + \frac{3}{14} + \frac{2}{7} = \frac{2+3+4}{14} = \boxed{\frac{9}{14}}$

$P(Y \le 2 | X=2) = f_{Y|X}(1|2) + f_{Y|X}(2|2) = \frac{3}{18} + \frac{4}{18} = \boxed{\frac{7}{18}}$

$P(X=2 | Y=3) = f_{X|Y}(2|3) = \boxed{\frac{2}{9}}$

6) 1. $f_Y(1) = \frac{1}{2}$ $f_Y(2) = \frac{1}{2}$ $f_X(1) = \frac{1}{2}$ $f_X(2) = \frac{1}{2}$

$f_Y(y) = \frac{1}{2}$ $f_X(x) = \frac{1}{2}$

$y=1$: $f_{X|Y}(1|1) = \frac{\frac{3}{8}}{\frac{1}{2}} = \boxed{\frac{3}{4}}$ $f_{X|Y}(2|1) = \frac{\frac{1}{8}}{\frac{1}{2}} = \boxed{\frac{1}{4}}$

$y=2$: $f_{X|Y}(1|2) = \frac{\frac{1}{8}}{\frac{1}{2}} = \boxed{\frac{1}{4}}$ $f_{X|Y}(2|2) = \frac{\frac{3}{8}}{\frac{1}{2}} = \boxed{\frac{3}{4}}$

$x=1$: $f_{Y|X}(1|1) = \frac{\frac{3}{8}}{\frac{1}{2}} = \boxed{\frac{3}{4}}$ $f_{Y|X}(2|1) = \frac{\frac{1}{8}}{\frac{1}{2}} = \boxed{\frac{1}{4}}$

$x=2 : p_{Y|X}(1|2) = \frac{\frac{1}{8}}{\frac{1}{2}} = \boxed{\frac{1}{4}}$ $f_{Y|X}(2|2) = \frac{\frac{3}{8}}{\frac{1}{2}} = \boxed{\frac{3}{4}}$

$$f_{X|Y}(x|1) = \begin{cases} \frac{3}{4}, & x=1 \\ \frac{1}{4}, & x=2 \end{cases} \qquad f_{X|Y}(x|2) = \begin{cases} \frac{1}{4}, & x=1 \\ \frac{3}{4}, & x=2 \end{cases}$$

$$f_{Y|X}(y|1) = \begin{cases} \frac{3}{4}, & y=1 \\ \frac{1}{4}, & y=2 \end{cases} \qquad f_{Y|X}(y|2) = \begin{cases} \frac{1}{4}, & y=1 \\ \frac{3}{4}, & y=2 \end{cases}$$

Means:

$E[X|Y=1] = \sum_{x \in S_X} x f(x|1) = 1 \cdot \frac{3}{4} + 2 \cdot \frac{1}{4} = \boxed{\frac{5}{4}}$ $M_{X|1}$

$E[X|Y=2] = \sum_{x \in S_X} x f(x|2) = 1 \cdot \frac{1}{4} + 2 \cdot \frac{3}{4} = \boxed{\frac{7}{4}}$ $M_{X|2}$

$E[Y|X=1] = \sum_{y \in S_Y} y f(y|1) = 1 \cdot \frac{3}{4} + 2 \cdot \frac{1}{4} = \boxed{\frac{5}{4}}$ $M_{Y|1}$

$E[Y|X=2] = \sum_{y \in S_Y} y f(y|2) = 1 \cdot \frac{1}{4} + 2 \cdot \frac{3}{4} = \boxed{\frac{7}{4}}$ $M_{Y|2}$

Variances

$\sigma^2_{X|Y} = E[X^2|Y] - (M_{X|Y})^2$

$E[X^2|Y=1] = 1^2(\frac{3}{4}) + 2^2(\frac{1}{4}) = \frac{3}{4} + 1 = \boxed{\frac{7}{4}}$ $\frac{7}{4} - (\frac{5}{4})^2 = \frac{7}{4} - \frac{25}{16} = \boxed{\frac{3}{16}}$ $\sigma^2_{X|1}$

$E[X^2|Y=2] = 1^2(\frac{1}{4}) + 2^2(\frac{3}{4}) = \frac{1}{4} + 3 = \boxed{\frac{13}{4}}$ $\frac{13}{4} - (\frac{7}{4})^2 = \frac{52}{16} - \frac{49}{16} = \boxed{\frac{3}{16}}$ $\sigma^2_{X|2}$

$E[Y|X=1] = 1^2(\frac{3}{4}) + 2^2(\frac{1}{4}) = \frac{3}{4} + 1 = \boxed{\frac{7}{4}}$ $\frac{7}{4} - (\frac{5}{4})^2 = \boxed{\frac{3}{16}}$ $\sigma^2_{Y|1}$

$E[Y|X=2] = 1^2(\frac{1}{4}) + 2^2(\frac{3}{4}) = \boxed{\frac{13}{4}}$ $\frac{13}{4} - (\frac{7}{4})^2 = \boxed{\frac{3}{16}}$ $\sigma^2_{Y|2}$